

Technical Report: AI Room Capture

Applied AI Engineer Assessment — Multi-tier spatial reconstruction pipeline

1. Problem and Objective

The objective of AI Room Capture is to build a spatial reconstruction pipeline that turns consumer device captures into structured property data. The assessment requires three capture tiers: unordered photographs, handheld RGB video, and LiDAR-enabled capture. The intended output includes dimensioned room plans, ceiling height, floor area, openings, multi-room adjacency, damage information, scope line items, and confidence intervals.

I treated the project first as a systems problem rather than starting with a large reconstruction model. The first goal was to establish a common input/output contract, a single capture interface, reproducible execution, and a working baseline for at least one capture tier. This also makes it possible to evaluate later reconstruction models without changing the downstream data flow.

2. System Approach

The pipeline uses one command-line entry point: `python -m src.run --tier ... --input ...`. The tier determines which runtime path is used, while the output is standardized into a common JSON representation. This keeps capture-specific processing separate from later geometry, semantic and damage stages.

The current architecture is intentionally modular. Tier 1 has a functional computer-vision baseline. Tier 2 and Tier 3 currently have runtime adapters that validate and describe captures. The intended next layer is stronger reconstruction followed by global optimization, semantic vectorization, damage localization, and uncertainty propagation.

3. Capture Tiers and Common Output

Tier	Input	Current state	Intended direction
1 — Photos	2–8 unordered stills	ORB matching, RANSAC verification, development geometry baseline	Global by sensor multi-view reconstruction
2 — Video	Handheld RGB walkthrough	Input validation and metadata extraction	Visual tracking + depth + reconstruction
3 — LiDAR	RGB + depth + poses + intrinsics	Capture-directory validation and metadata	Sensor-aware RGB/depth reconstruction

Regardless of capture tier, the output contract includes room dimensions, uncertainty intervals, openings, surfaces, damage fields, concealed-damage fields, scope, confidence and provenance. A rendered room plan is also produced for the current room baseline.

4. What I Implemented

For Tier 1, I implemented image loading, ORB feature extraction, feature matching and RANSAC-based geometric verification. ORB was selected for the first iteration because it is lightweight and directly available through OpenCV. It was sufficient to establish whether the unordered-image pipeline was finding repeatable correspondences before introducing a heavier reconstruction model.

On the development three-image fixture, `room_1.jpg` to `room_2.jpg` produced 288 good matches and 198 geometric inliers (68.8%). `room_2.jpg` to `room_3.jpg` produced 288 good matches and 194 inliers (67.4%). The pipeline then generated a development room estimate, opening candidates, `result.json`, and `room_plan.png`.

The current geometry estimator is deliberately labeled as a development baseline. Opening dimensions are still uncalibrated, so the output is not presented as metric benchmark evidence.

For Tier 2, the runtime adapter validates the video and records frame count, FPS, resolution and duration. For Tier 3, the adapter validates the capture directory, enumerates available files and records metadata. These adapters establish the execution route, but they are not being represented as complete 3D reconstruction systems.

5. Evaluation and Evidence

I added a reproducible benchmark evaluator at `scripts/benchmark/evaluate.py`. It accepts prediction and ground-truth JSON files and reports absolute error, percentage error, metric-specific tolerance, pass/fail and an aggregate pass rate.

The repository currently does not contain physical laser/tape ground truth or a real benchmark capture set for this implementation. The synthetic fixture is therefore used only to verify execution, matching, output generation and schema behavior. I did not use it to claim benchmark accuracy.

6. Fix Loop

The assessment requires a measurable fix loop. I captured before and after artifacts for an attempted opening-detection refinement. The proposed change used additional geometric and edge-support information to reduce weak rectangular candidates.

The result was negative: the development fixture produced the same two opening detections before and after the attempted refinement. I kept the negative result rather than manufacturing an improvement. The before/after JSON and rendered plans are stored under `reports/fix_loop/` and the outcome is documented in `diff.md`.

The result suggests that local 2D filtering is not the main bottleneck. A more useful next step is stable multi-view geometry and metric calibration, followed by opening refinement on the reconstructed surfaces.

7. Limitations and Next Steps

- **Metric reconstruction:** the current room estimator provides a development estimate and does not establish absolute scale from real captures.
- **Multi-room reconstruction:** room stitching, adjacency recovery and drift correction are not yet implemented.
- **Video and LiDAR:** the current implementations validate capture/runtime inputs rather than performing full reconstruction.
- **Damage:** damage fields exist in the output contract, but full segmentation and 3D registration remain to be implemented.
- **Benchmark:** real captures and laser/tape measurements are required before the specified accuracy gates can be reported.

The planned reconstruction path is to replace the Tier 1 baseline with a pose-free wide-baseline model such as MAST3R, use visual tracking and monocular depth for Tier 2, and use raw RGB/depth/pose information directly for Tier 3. A global pose/factor graph with loop closure and structural constraints would then make the geometry consistent across rooms. Semantic vectorization would separate walls, ceilings and openings, while 2D damage segmentation could be registered back to reconstructed surfaces. Calibration and detection uncertainty would then be propagated into the final measurement intervals.

8. Conclusion

The main result of this iteration is an end-to-end, reproducible foundation rather than a claim of completed production reconstruction. The repository has a single capture interface, a working photo baseline, runtime paths for all three tiers, a shared spatial output schema, benchmark evaluation infrastructure, documentation and a recorded fix loop. The remaining work is concentrated in metric reconstruction, global optimization, multi-room stitching, semantic/damage processing and real benchmark validation.

If I continued the project, I would first collect the real benchmark captures and ground-truth measurements, establish the current metric error, and then improve the reconstruction stage against that measured baseline. This would give each later model or optimization change a concrete failure mode to target.

Evidence note: Development results in this report are explicitly separated from benchmark evidence. No benchmark accuracy gate is claimed as passed without the required real-world ground truth.